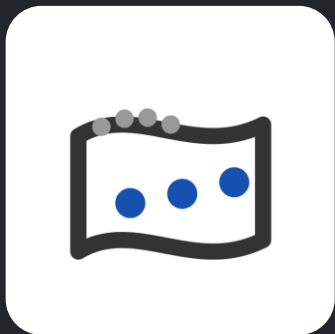




Simplify surface

Rebuild a heavy face as a lighter cubic B-spline within a tolerance -- or replace it in place.

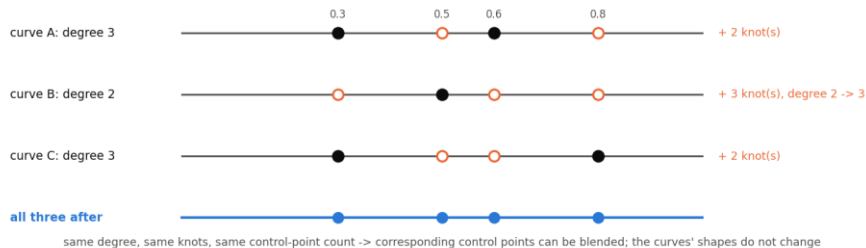


What it does

Imported, derived and fitted faces carry far more control points than their shape needs, which slows and destabilises later features. Simplify surface finds, curve by curve, the fewest control points within your tolerance.

Use it when

- A heavy imported or fitted face
- Downstream features are slow or fail on a face
- Replace a face with a lighter one in place



Station curves are reduced, made compatible and skinned.



The dialog

IMAGE MISSING:
docs/decks/simplify_surface/shots/dialog.png

#	Parameter	What it means
1	Face	The face to simplify.
2	Tolerance	0.01-100 mm (1 mm).
3	Continuity	G0 / G1 / G2 end derivatives along v (G2 mode has no effect).
4	Mode	AUTO (Greville stations, binary search per curve) / MANUAL (U, V counts).
5	Replace face	Replace the source face instead of a new sheet.
6	Debug print	Console.



Outputs and messages

Outputs

Key	What it holds
simplified	The new sheet (or the replaced face).

Messages

	Message / what to do
(none)	



Tips and limits



Checked on stations

The error is measured along the station curves only.



Compatibility re-inflates

The final count follows the most complex station; G1 needs about 6 control points.



Untested

No test fixture yet.



Where it lives

Document	<code>gordonSurface (d0950ed72a894fbe35c27d43)</code>
Tab	<code>simplifySurface</code>
Tests	none yet
Examples	Test Part Studio 1: Simplify surface 1
Explainer	<code>docs/explainers/gordon_surface/gordon_surface_explained.md</code> , section 2.6
Related	Gordon Surface, Interior curves, Scaled Curve, Modify curve end, Pull surface, Simplify surface